William Hadden

Sydney, Australia | williamhadden92@gmail.com | (+61) 493 189 782 | New Zealand Citizen

PhD researcher passionate about the intersection of machine learning and geometric analysis.

EDUCATION

Current PhD Student, Machine Learning Methods in Geometric Analysis **Sydney, Australia**
School of Pure Mathematics at University of New South Wales **Feb 2025 - Present**

My research explores machine learning assisted discovery of solutions to non-linear geometric PDEs, combining numerical approximation with rigorous mathematical verification.

Publications:

- The Kähler-Einstein metric on the third del Pezzo surface (joint project with group from Imperial College London), awaiting publication (https://arxiv.org/abs/2609.32655)

Notable achievements:

- February 2025: Awarded Research Training Program scholarship (3.5 years funded position), competitively awarded
- December 2025: Invited presenter at the Annual Meeting of the Australian Mathematical Society (only first year UNSW PhD student invited to speak)
- February 2026: Invited speaker at the Australian Mathematical Research Institute's workshop in Analytic and Geometric Methods on Complex Manifolds (only UNSW PhD student invited)
- OzGeomPDE: Organizing member of a week long research intensive for Australian PhD students working in PDEs at the MATRIX research institute in Victoria (competitively awarded, one per year in Australia)

Teaching:

- Casual Academic at UNSW, tutored Mathematics 1A & 1B, Higher Mathematics 1B, Topology and differential geometry, Higher Topology and Differential Geometry

Other Activities

- July 2026, AMSI Winter School in Geometric Analysis

Masters in Pure Mathematics **Sydney, Australia**
School of Pure Mathematics at University of New South Wales **2023-2024**
Thesis Grade: High Distinction

- Research on computationally assisted proofs for symmetric solutions of Ricci flow using centre manifold theory

Bachelor of Arts and Science **Dunedin, New Zealand**
University of Otago **2019-2022**
Majors: Chemistry, Computer Science; Minor: Mathematics; GPA: 8.7/9.0

Notable Achievements

- Proudfoot scholarship in Experimental Science (2022), awarded to the top third year student across all the experimental sciences
- University of Otago Council Commendation for Exceptional Academic Performance

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Part time) **Sydney, Australia**
Image Analyzer **Jul '22 - Jul '25**

- Image Analyzer is a New Zealand based AI start-up that detects visual threats with unique artificial intelligence-based content moderation software. I worked part-time throughout my undergraduate and master's degrees in developing machine learning models for content moderation, including sentiment analysis and computer vision systems detecting harmful imagery

Machine Learning Researcher (Intern) **Sydney, Australia**
MongoDB **Nov '23 – Feb '24**

- Developed structure-preserving tokenization methods for nested JSON databases
- Built ML models for density estimation of semi-structured data

Technology Consultant (Intern) **Auckland, New Zealand**
PWC **Nov '21 - Feb '22**

- Contributed to the automation of the national COVID 19 isolation program

Computational Modelling Researcher (Intern) **Dunedin, New Zealand**
AbacusBio **Nov '20 - Mar '21**

- Developed & researched mathematical models for honeybee population dynamics.
- Implemented these models in python to assess viability of commercial beekeeping operations in remote parts of Asia

Infantry Solider (Full time and Reserves) **Dunedin, New Zealand**
New Zealand Defense Force **Nov '19 – Sep '22**

- Completed four months of full time, high intensity basic training. Served part-time in the 2/4 Infantry Regiment while keeping up university course commitments

TECHNICAL SKILLS

- **Mathematics:** Differential & Riemannian Geometry, Nonlinear PDEs, Functional Analysis, Geometric Measure Theory, Topology and Modules & Representations
- **Machine Learning:** Research and Development of generative AI systems for applications in semi-structured data analysis & vision models
- **Data Science:** Skilled in data analysis and aggregation pipelines, especially for ML applications.
- **Programming:** Skilled programming ability in Python & Java, familiarity with C languages, Kotlin and common web development frameworks like React. Extensive experience with AWS in machine learning projects

LEISURE INTERESTS

- **Outdoor pursuits:** Surfing, trail running, hiking, climbing, rugby union, alpine skiing
- **Intellectual interests:** Current events, geopolitics, economics, history and science
- **Music:** Classical Piano, Jazz and Blues

- **Volunteering:** Volunteer tutor students at Weave Youth Services who come from low-socioeconomic backgrounds and are primarily of aboriginal descent. In addition to academic support, I act as a "big brother" mentor.